Medvale Internal Medicine

Chapter 3B: Esophagus and Stomach — GERD, Dysphagia, Peptic Ulcer Disease, Gastritis, and Upper GI Bleeding

Original Medvale Study Resource

Chapter purpose

This chapter is original Medvale writing. The uploaded GI reference was used only as a coverage checklist for esophageal and stomach topics: esophageal cancer, achalasia, diffuse esophageal spasm, hiatal hernia, Mallory-Weiss syndrome, Plummer-Vinson syndrome, Schatzki ring, caustic injury, esophageal diverticula, esophageal perforation, peptic ulcer disease, gastritis, *H. pylori*, and ulcer complications. No source images or source wording are reproduced.

Esophageal and gastroduodenal disease is high-yield because it sits between routine outpatient care and emergencies. A patient with uncomplicated burning epigastric pain may need empiric acid suppression and *H. pylori* testing. A patient with progressive dysphagia, anemia, bleeding, weight loss, or persistent vomiting needs endoscopic evaluation. A patient with hematemesis needs resuscitation before diagnostic refinement. A patient with severe chest pain after vomiting may have a transmural esophageal rupture and must be treated as a surgical emergency.

The organizing principle is to separate disease mechanisms. Reflux disease is chemical injury from gastric contents. Dysphagia is either a transit problem or a narrowed lumen. Peptic ulcer disease is a breach in mucosal defense, usually from *H. pylori* or NSAIDs. Upper GI bleeding is a hemodynamic problem first and an endoscopic problem second. Esophageal perforation is a source-control emergency because contamination of the mediastinum or pleural space can rapidly produce sepsis.

Core illness scripts

Problem	Typical script	Key test	Management priority
GERD	Postprandial heartburn and regurgitation, often worse lying down. May cause chronic cough or hoarseness.	Empiric PPI if classic symptoms and no alarm features. EGD if alarm features or refractory disease.	Lifestyle measures, acid suppression, and evaluation for complications when indicated.
Barrett esophagus	Chronic GERD risk profile with intestinal metaplasia of distal esophagus. Symptoms may be absent.	EGD with biopsy.	Acid suppression plus surveillance or endoscopic eradication depending on dysplasia.
Esophageal cancer	Progressive dysphagia to solids, then liquids; weight loss, anemia, bleeding, aspiration, or hoarseness.	EGD with biopsy, then staging.	Tissue diagnosis and staging before treatment planning.
Achalasia	Dysphagia to solids and liquids, regurgitation of undigested food, nocturnal aspiration, weight loss.	Manometry confirms; EGD rules out pseudoachalasia.	LES-directed therapy such as pneumatic dilation, Heller myotomy, POEM, or botulinum toxin in selected patients.
Diffuse esophageal spasm	Intermittent dysphagia with noncardiac chest pain that may mimic angina.	High-resolution manometry.	Exclude cardiac disease when appropriate; symptom-directed smooth-muscle relaxation or neuromodulation.
Mallory-Weiss tear	Hematemesis after retching or vomiting; usually mucosal and self-limited.	EGD if ongoing bleeding or diagnostic uncertainty.	Resuscitate if needed; endoscopic hemostasis if bleeding persists.
Boerhaave syndrome	Severe chest or epigastric pain after vomiting, fever, tachycardia, pleural effusion, mediastinal air, sepsis.	Water-soluble contrast esophagram or CT esophagram.	NPO, broad antibiotics, IV fluids, drainage/source control, urgent surgical or endoscopic repair.

Problem	Typical script	Key test	Management priority
Peptic ulcer disease	Epigastric burning or gnawing pain with *H. pylori*, NSAID, or stress exposure.	EGD for alarm features or bleeding; noninvasive *H. pylori* testing in selected uncomplicated dyspepsia.	Remove cause, suppress acid, eradicate *H. pylori* when present.
Upper GI bleeding	Hematemesis, coffee-ground emesis, melena, syncope, anemia, or shock.	EGD after stabilization.	Airway/circulation first, risk stratification, targeted therapy, endoscopic hemostasis.

Approach to dysphagia

Dysphagia means impaired passage of food or liquid from the mouth to the stomach. The first split is oropharyngeal versus esophageal. Oropharyngeal dysphagia causes difficulty initiating a swallow, coughing, choking, nasal regurgitation, aspiration, or a wet voice after swallowing. It often reflects neurologic disease, structural head and neck disease, or impaired coordination. Esophageal dysphagia feels as if food gets stuck after swallowing.

The second split is mechanical obstruction versus motility failure. Mechanical obstruction usually affects solids before liquids because liquids can still pass through a narrowed lumen. Motility disorders affect solids and liquids early because the transport mechanism itself is abnormal. A slowly progressive course is more concerning than an intermittent one. Progressive solid-food dysphagia with weight loss or anemia is esophageal cancer until proven otherwise. Intermittent solid-food dysphagia suggests Schatzki ring, eosinophilic esophagitis, or intermittent food impaction. Dysphagia to solids and liquids from the beginning suggests achalasia, diffuse esophageal spasm, or scleroderma esophagus.

Original dysphagia pattern diagram

```
                          DYSPHAGIA
                             |
            +----------------+----------------+
            |                                 |
  Trouble initiating swallow       Food sticks after swallow
  cough/choke/aspiration           = esophageal dysphagia
  = oropharyngeal pattern                  |
                                           |
                   +-----------------------+-----------------------+
                   |                                               |
             Solids first                                Solids + liquids early
         mechanical obstruction                           motility disorder
                   |                                               |
      +------------+------------+                 +----------------+----------------+
      |                         |                 |                                 |
 Intermittent              Progressive        Regurgitation/weight loss        Chest pain prominent
 ring, web, EoE            cancer/stricture   achalasia/pseudoachalasia        esophageal spasm
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A barium esophagram is useful when a structural map is needed, when a diverticulum is suspected, or when perforation risk changes the choice of test. Endoscopy is essential when alarm features exist because tissue diagnosis may be needed. Manometry is the physiologic test for suspected motility disease after structural causes have been excluded.

GERD, Barrett esophagus, and reflux complications

Gastroesophageal reflux disease occurs when reflux of gastric contents produces troublesome symptoms or mucosal injury. Typical symptoms are heartburn and regurgitation. Heartburn is retrosternal burning that often occurs after meals or when lying down. Regurgitation is effortless return of gastric contents into the throat or mouth. Symptoms may worsen with large meals, late meals, obesity, pregnancy, hiatal hernia, alcohol, tobacco, peppermint, chocolate, or medications that lower lower esophageal sphincter tone.

Extraesophageal associations include chronic cough, laryngitis, hoarseness, asthma-like symptoms, dental enamel injury, and noncardiac chest discomfort. These associations are less specific than classic heartburn and regurgitation. A patient with isolated cough should not automatically be labeled as reflux-related without considering asthma, upper airway cough syndrome, ACE-inhibitor effect, smoking-related disease, and malignancy risk.

In a low-risk patient with classic symptoms and no alarm features, an empiric proton pump inhibitor trial is reasonable. A typical trial is about 8 weeks, with the PPI taken before the first meal of the day. Alarm features should shift the approach to upper endoscopy: dysphagia, odynophagia, GI bleeding, iron deficiency anemia, weight loss, persistent vomiting, or concern for malignancy. Refractory symptoms may require checking adherence and timing, escalating acid suppression, endoscopy, ambulatory reflux monitoring, or evaluation for non-GERD causes.

Barrett esophagus is intestinal metaplasia of the distal esophagus caused by chronic reflux injury. Its importance is cancer prevention, because Barrett esophagus increases the risk of esophageal adenocarcinoma. Risk is enriched by chronic GERD, male sex, age over 50, central obesity, tobacco exposure, White race in U.S. epidemiology, and family history. Barrett esophagus itself may not cause symptoms. Endoscopic surveillance and treatment depend on biopsy histology: nondysplastic Barrett is usually surveilled, while dysplasia can lead to endoscopic eradication therapy.

GERD-related entity	Mechanism	Clinical clue	Test or management anchor
Nonerosive reflux disease	Reflux symptoms without visible erosions	Heartburn/regurgitation, normal EGD possible	Empiric PPI if no alarm signs; reflux monitoring if diagnosis uncertain.
Erosive esophagitis	Acid/pepsin injury causes visible mucosal breaks	Heartburn, odynophagia, bleeding in severe disease	PPI therapy; EGD documents severity.
Peptic stricture	Chronic inflammation scars distal esophagus	Progressive solid dysphagia with GERD history	EGD to exclude cancer; dilation plus acid suppression.
Barrett esophagus	Intestinal metaplasia from chronic reflux	Often asymptomatic; risk factor profile matters	EGD with biopsies; surveillance or eradication depending on dysplasia.
Adenocarcinoma	Dysplasia-carcinoma sequence in distal esophagus/GE junction	Progressive dysphagia, weight loss, anemia	EGD biopsy and staging.

Esophageal cancer

Esophageal cancer presents late because the esophageal lumen can narrow substantially before symptoms become obvious. The classic symptom is progressive dysphagia that begins with solids and advances to liquids. Weight loss is common because eating becomes difficult and malignancy creates systemic catabolism. Odynophagia, anemia, hematemesis, aspiration pneumonia, hoarseness from recurrent laryngeal nerve involvement, and chest pain suggest advanced local disease or complications.

Squamous cell carcinoma is associated with tobacco, alcohol, caustic injury, achalasia, prior radiation, Plummer-Vinson syndrome, and certain dietary or environmental exposures. It more often involves the upper or mid esophagus. Adenocarcinoma is linked most strongly with chronic GERD, Barrett esophagus, central obesity, and distal esophageal or gastroesophageal junction disease. Tobacco is a risk factor for both major histologies, while alcohol is more strongly linked to squamous cell carcinoma than adenocarcinoma.

Diagnosis requires upper endoscopy with biopsy. Barium swallow may show a stricture or shouldering lesion but cannot replace tissue diagnosis. Staging commonly uses CT for regional or distant spread, endoscopic ultrasound for depth and nodal assessment in potentially resectable disease, and PET/CT in selected pathways. Treatment depends on stage, location, histology, and fitness for therapy. Very early mucosal disease may be treated endoscopically; locoregional disease often requires combined oncology and surgical planning; advanced obstructive disease may require palliation with stenting, radiation, systemic therapy, or feeding access.

Feature	Squamous cell carcinoma	Adenocarcinoma
Typical location	Upper or middle esophagus	Distal esophagus or gastroesophageal junction
Major risks	Tobacco, alcohol, caustic injury, achalasia, Plummer-Vinson syndrome, prior radiation	Chronic GERD, Barrett esophagus, central obesity, tobacco
Typical presentation	Progressive dysphagia, weight loss, odynophagia, aspiration, hoarseness if advanced	Progressive dysphagia and weight loss, often with GERD/Barrett background
Diagnostic anchor	EGD with biopsy	EGD with biopsy
Treatment logic	Stage first; local therapy only for early disease; combined modalities common	Stage first; endoscopic, surgical, systemic, and radiation options depend on depth/nodes

Major esophageal motility disorders

Achalasia is failure of lower esophageal sphincter relaxation plus loss of coordinated esophageal peristalsis. Food and liquid enter the esophagus but cannot reliably traverse the gastroesophageal junction. Patients develop dysphagia to solids and liquids, regurgitation of undigested food, chest discomfort, weight loss, halitosis, and aspiration events. Symptoms may worsen when eating quickly or lying down. Barium swallow may show distal tapering with proximal dilation, but high-resolution manometry is the diagnostic standard. Endoscopy is still needed because pseudoachalasia from malignancy can mimic idiopathic achalasia, especially in older patients with rapid weight loss or short symptom duration.

Therapy lowers lower-esophageal-sphincter resistance. Pneumatic dilation disrupts LES muscle fibers. Laparoscopic Heller myotomy and peroral endoscopic myotomy cut the muscle layer to permit emptying. Botulinum toxin injection can help patients who are poor candidates for more durable procedures, but benefit is often temporary. Medical smooth-muscle relaxants such as nitrates or calcium-channel blockers are less effective and are generally reserved for limited situations.

Diffuse esophageal spasm produces uncoordinated premature contractions that cause intermittent dysphagia and noncardiac chest pain. The lower esophageal sphincter relaxes normally, which distinguishes it from achalasia. Barium swallow may show a corkscrew pattern, but high-resolution manometry is the key diagnostic test. Treatment is symptom-directed and may include calcium-channel blockers, nitrates, peppermint oil, neuromodulators, or selected procedural therapy in severe refractory disease. Because the chest pain can mimic angina, ischemic heart disease should be excluded when risk factors or symptoms warrant.

Scleroderma esophagus is the opposite lower-sphincter problem from achalasia. Smooth muscle atrophy and fibrosis cause weak peristalsis and a hypotensive lower esophageal sphincter. The result is severe reflux, dysphagia, and aspiration risk. The graph relationship is important: achalasia is high LES pressure with poor relaxation; scleroderma is low LES pressure with poor clearance.

Disorder	Mechanism	Classic presentation	Best diagnostic concept	Treatment concept
Achalasia	LES fails to relax; esophageal aperistalsis	Solids + liquids dysphagia, regurgitated undigested food, aspiration, weight loss	Manometry confirms; EGD excludes pseudoachalasia	Pneumatic dilation, Heller myotomy, POEM, botulinum toxin in selected patients
Diffuse esophageal spasm	Premature, nonperistaltic contractions; LES relaxation preserved	Intermittent dysphagia plus chest pain that may mimic angina	High-resolution manometry; barium may show corkscrew pattern	Smooth-muscle relaxants and symptom control; procedural therapy rarely

Disorder	Mechanism	Classic presentation	Best diagnostic concept	Treatment concept
Scleroderma esophagus	Weak peristalsis plus low LES tone	Severe reflux plus dysphagia and aspiration risk	Manometry shows poor peristalsis and hypotensive LES	Aggressive reflux control and aspiration precautions

Rings, webs, diverticula, and hiatal hernia

Schatzki ring is a thin distal esophageal ring that causes intermittent solid-food dysphagia, often with meat or bread. It is commonly linked to sliding hiatal hernia. Symptoms are episodic because the lumen is narrowed but not progressively invaded. Symptomatic rings are treated with dilation and reflux control when reflux coexists.

Esophageal webs are more proximal. Plummer-Vinson syndrome combines upper esophageal webs, iron deficiency anemia, glossitis or atrophic mucosa, and increased risk of squamous cell carcinoma of the upper aerodigestive tract. The management principle is to correct iron deficiency, evaluate dysphagia, and dilate when needed. The cancer risk association matters because this is not merely a benign swallowing complaint.

Esophageal diverticula are outpouchings that often reflect an underlying pressure or motility problem. Zenker diverticulum arises above the upper esophageal sphincter because of cricopharyngeal dysfunction. It causes regurgitation of undigested food, halitosis, cough, aspiration, and sometimes a neck gurgle. Epiphrenic diverticula occur near the distal esophagus and are often associated with motility disorders. Barium esophagram is often the safest first test when a diverticulum is suspected because it maps anatomy before instrumentation.

Hiatal hernia means part of the stomach has moved upward through the diaphragmatic hiatus. Type I sliding hernia is most common and is strongly associated with GERD because the gastroesophageal junction moves above the diaphragm. Paraesophageal hernias are less common but more concerning because the gastroesophageal junction may remain near normal position while the fundus herniates. Obstruction, volvulus, bleeding, incarceration, and strangulation drive surgical consideration. The exam clue is that sliding hernia behaves like reflux, while paraesophageal hernia behaves like an anatomic mechanical risk.

Esophageal injury: Mallory-Weiss, Boerhaave, and caustic ingestion

Mallory-Weiss syndrome is a mucosal laceration at or near the gastroesophageal junction after forceful retching, vomiting, coughing, seizures, or sudden intra-abdominal pressure increases. It causes hematemesis ranging from blood streaking to significant bleeding. Most cases stop spontaneously. Persistent bleeding can be treated endoscopically with clips, injection, cautery, or banding depending on the lesion. Because the lesion is mucosal, systemic toxicity and mediastinitis are not expected unless another injury exists.

Boerhaave syndrome is transmural esophageal rupture, classically after violent vomiting. It is a surgical emergency because swallowed contents enter the mediastinum or pleural space, causing mediastinitis, sepsis, pleural effusion, pneumothorax, or shock. The classic triad of vomiting, chest pain, and subcutaneous emphysema is helpful but insensitive. Management is NPO, IV fluids, broad-spectrum antibiotics, acid suppression, drainage/source control, and urgent surgical or endoscopic repair depending on stability, location, containment, and timing. Delays beyond the first 24 hours substantially worsen outcomes.

Caustic ingestion injures the esophagus by chemical necrosis. Alkali ingestion can liquefy tissue and penetrate deeply; acid ingestion more often causes coagulative injury but can still be devastating. Do not induce vomiting, perform blind gastric lavage, or attempt neutralization because these actions can worsen injury. Early airway assessment is critical, especially with drooling, stridor, or oropharyngeal burns. Endoscopic grading is usually performed within the first 12 to 24 hours in significant ingestions if the patient is stable and there is no perforation. Late complications include strictures and increased cancer risk after severe injury.

Problem distinction: vomiting followed by hematemesis with stable appearance suggests Mallory-Weiss tear. Vomiting followed by severe chest pain, fever, tachycardia, pleural effusion, or mediastinal air suggests Boerhaave syndrome until proven otherwise.

Peptic ulcer disease and gastritis

Peptic ulcer disease is mucosal ulceration caused by an imbalance between injurious forces and mucosal defense. The two dominant causes are *Helicobacter pylori* infection and NSAID exposure. *H. pylori* promotes chronic gastritis, disrupts mucosal protection, and can increase acid exposure in the duodenum. NSAIDs inhibit prostaglandin synthesis, decreasing mucous and bicarbonate secretion and reducing mucosal blood flow. Physiologic stress in critical illness, severe burns, head injury, mechanical ventilation, and coagulopathy can predispose to stress-related mucosal disease.

Symptoms are not perfectly reliable. Duodenal ulcers often cause epigastric burning or gnawing pain that may improve temporarily with meals and recur hours later or at night. Gastric ulcers can worsen with meals and may produce early satiety, nausea, or weight loss, but overlap is common. Because gastric cancer can mimic a benign gastric ulcer, gastric ulcers seen on endoscopy are biopsied. Duodenal ulcers are less likely malignant, but complicated or atypical cases still require thoughtful evaluation. Dyspepsia with alarm features, older age at onset, GI bleeding, anemia, progressive vomiting, dysphagia, or weight loss requires endoscopy rather than prolonged empiric treatment.

Gastritis is mucosal inflammation that may be acute or chronic. Common triggers include *H. pylori*, NSAIDs, alcohol, bile reflux, severe physiologic stress, and autoimmune gastritis. Symptoms can include epigastric discomfort, nausea, vomiting, early satiety, or bleeding, but histologic gastritis and symptoms do not always correlate. Autoimmune gastritis involves parietal cell loss, achlorhydria, intrinsic factor deficiency, vitamin B12 deficiency, hypergastrinemia, and increased risk of gastric neuroendocrine tumors and adenocarcinoma.

Feature	Duodenal ulcer	Gastric ulcer
Common causes	*H. pylori* very common; NSAIDs also important	*H. pylori* and NSAIDs; malignancy must be considered
Meal relationship	Pain may improve with food and recur later; nocturnal pain is common	Pain may worsen with meals; nausea and early satiety may occur
Cancer concern	Low but not zero; biopsy if atypical or nonhealing	Higher concern; biopsy gastric ulcer to exclude malignancy
Complications	Bleeding, perforation, gastric outlet obstruction from duodenal bulb/pyloric edema or scarring	Bleeding, perforation, weight loss, malignancy mimic
Treatment core	Eradicate *H. pylori*, stop NSAID if possible, PPI	Biopsy if endoscoped, eradicate *H. pylori*, stop NSAID if possible, PPI

H. pylori testing and treatment framework

H. pylori should be tested when the result will change management: active or prior peptic ulcer disease, gastric MALT lymphoma, selected dyspepsia pathways, unexplained iron deficiency anemia after appropriate evaluation, immune thrombocytopenia in selected settings, and before long-term NSAID therapy in higher-risk patients. Tests for active infection include urea breath test, stool antigen test, and biopsy-based testing. Serology may remain positive after prior infection and is less useful for confirming active disease.

PPIs, bismuth, and antibiotics can cause false-negative active tests. When feasible, PPIs are held for about 2 weeks and antibiotics or bismuth for about 4 weeks before testing or test-of-cure. Treatment should reflect local resistance patterns and prior antibiotic exposure. When susceptibility is unknown, a current default regimen is optimized bismuth quadruple therapy for 14 days: PPI, bismuth, tetracycline, and metronidazole. Clarithromycin-based triple therapy should not be used empirically in many regions unless susceptibility is known or local resistance is very low. Proof of eradication is required after therapy using urea breath test, stool antigen, or biopsy-based testing.

Original H. pylori management loop

```
Indication to test
       |
       v
Active infection test: stool antigen, urea breath, or biopsy
       |
       +--> avoid false negatives: hold PPI about 2 weeks when feasible
       |                           hold antibiotics/bismuth about 4 weeks
```

```
       v
If positive: choose eradication regimen based on susceptibility and prior exposure
       |
       +--> if susceptibility unknown: optimized bismuth quadruple therapy x 14 days
       v
Confirm eradication with an active test after treatment and washout
```

Upper GI bleeding

Upper GI bleeding is bleeding proximal to the ligament of Treitz. Presentation includes hematemesis, coffee-ground emesis, melena, syncope, orthostasis, shock, or unexplained iron deficiency anemia. A high blood urea nitrogen relative to creatinine can support an upper source because digested blood protein increases urea absorption, but it is not diagnostic. Common etiologies include peptic ulcer disease, erosive gastritis or esophagitis, varices, Mallory-Weiss tear, malignancy, angioectasia, and Dieulafoy lesion. Variceal bleeding should be recognized early because it adds antibiotics, vasoactive therapy, and portal hypertension management to the general bleeding pathway.

Management begins before endoscopy. Assess airway protection, especially with massive hematemesis, altered mental status, or aspiration risk. Place two large-bore IVs or central access if needed, send CBC, CMP, coagulation studies, type and screen/crossmatch, and resuscitate with attention to perfusion and cardiopulmonary reserve. Restrictive transfusion is commonly used for stable nonvariceal bleeding, often at a hemoglobin threshold around 7 g/dL, while higher thresholds may be reasonable in active ischemia, severe ongoing bleeding, or individualized shock states. Hospitalized patients generally undergo upper endoscopy within 24 hours after presentation once stabilized.

Endoscopy identifies the bleeding lesion and provides hemostasis. High-risk ulcer stigmata include active spurting, active oozing, and a nonbleeding visible vessel. These require endoscopic therapy. Clean-based ulcers have low rebleeding risk and may allow earlier feeding and discharge depending on the overall scenario. After successful endoscopic therapy for high-risk ulcers, high-dose PPI therapy reduces rebleeding. If endoscopic therapy fails, interventional radiology embolization or surgery may be required.

Original upper GI bleeding pathway

```
Hematemesis, melena, syncope, anemia, or shock
       |
       v
Airway + circulation: mental status, aspiration risk, IV access, labs, type/screen
       |
       +--> Variceal clues? cirrhosis, portal hypertension, hematemesis, shock
       |         |-- yes: add vasoactive therapy + antibiotics + urgent endoscopic banding pathway
       |
       v
Risk stratify and resuscitate before endoscopy
       |
       v
EGD after stabilization, usually within 24 h if admitted/observed
       |
       +--> high-risk ulcer stigmata: endoscopic hemostasis + high-dose PPI
       +--> uncontrolled bleeding: repeat endoscopy, IR embolization, or surgery
```

Cause	Clue	Management logic
Peptic ulcer bleeding	Epigastric pain, NSAID or *H. pylori*, melena or hematemesis	Resuscitate, PPI, EGD with hemostasis for high-risk stigmata
Variceal bleeding	Cirrhosis, portal hypertension, large-volume hematemesis	Resuscitate, vasoactive drug, antibiotics, urgent EGD with banding
Mallory-Weiss tear	Hematemesis after retching; often self-limited	EGD if ongoing bleeding or diagnosis unclear; endoscopic therapy if persistent

Cause	Clue	Management logic
Erosive gastritis/esophagitis	NSAIDs, alcohol, critical illness, reflux symptoms	Acid suppression, remove trigger; EGD if significant bleeding or alarm features
Malignancy	Weight loss, anemia, progressive symptoms	EGD with biopsy; hemostasis may be temporary; stage and treat cancer

Complications of peptic ulcer disease

Bleeding is the most common serious complication of peptic ulcer disease and may present with melena, hematemesis, syncope, or iron deficiency anemia. Perforation causes sudden severe epigastric pain, peritonitis, free air under the diaphragm, and risk of sepsis. Management is NPO, IV fluids, broad antibiotics, acid suppression, surgical consultation, and source control. Gastric outlet obstruction produces early satiety, postprandial vomiting, dehydration, hypokalemic hypochloremic metabolic alkalosis, and a succussion splash. Causes include acute edema/inflammation or chronic scarring at the pylorus or duodenal bulb. Penetration occurs when an ulcer erodes into adjacent organs such as the pancreas, causing back pain or pancreatitis-like symptoms.

Complication	Clinical pattern	Diagnostic clue	Management priority
Bleeding	Melena, hematemesis, orthostasis, anemia	EGD identifies source and treats high-risk stigmata	Resuscitation, PPI, endoscopic hemostasis; IR/surgery if refractory
Perforation	Sudden severe pain, rigid abdomen, sepsis risk	Free air on upright CXR or CT	NPO, fluids, antibiotics, urgent surgical evaluation
Gastric outlet obstruction	Early satiety, nonbilious vomiting, weight loss, alkalosis	EGD/CT; retained food; pyloric or duodenal narrowing	NG decompression, fluids/electrolytes, ulcer treatment; dilation/surgery if fixed scar
Penetration	Pain radiates to back, persistent despite therapy	CT/endoscopy depending on suspected organ	Treat ulcer plus manage invaded-organ complication

High-yield numeric anchors

Anchor	Why it matters
Empiric GERD PPI trial: about 8 weeks	Appropriate only when symptoms are classic and alarm features are absent.
Alarm features	Dysphagia, odynophagia, weight loss, GI bleeding, persistent vomiting, or iron deficiency anemia push toward EGD.
Dysphagia to solids first	Mechanical obstruction until proven otherwise.
Dysphagia to solids and liquids from onset	Motility disorder until proven otherwise.
H. pylori active testing	PPI can cause false negatives; hold about 2 weeks when feasible.
Antibiotic/bismuth washout	Hold about 4 weeks when feasible before *H. pylori* active testing or test-of-cure.
Bismuth quadruple therapy	Common modern empiric choice for *H. pylori* when susceptibility is unknown; usually 14 days.
UGIB transfusion anchor	Hemoglobin threshold around 7 g/dL is used in many stable hospitalized patients; individualize for shock/ischemia.

Anchor	Why it matters
UGIB endoscopy timing	EGD generally within 24 hours after presentation for admitted/observed patients once stabilized.
Boerhaave source control	Delays beyond 24 hours worsen mortality and fistula risk.
Gastric ulcer biopsy	Gastric ulcers require malignancy exclusion.

Common pitfalls

1. **Treating progressive dysphagia as uncomplicated GERD.** Progressive solid-food dysphagia, weight loss, anemia, or bleeding deserves endoscopy. Acid suppression may reduce coexisting reflux symptoms but can delay cancer diagnosis if alarm features are ignored.
2. **Forgetting pseudoachalasia.** New achalasia-like symptoms in an older patient with rapid weight loss should prompt careful endoscopic and imaging evaluation for malignancy at the gastroesophageal junction.
3. **Confusing Mallory-Weiss tear with Boerhaave syndrome.** Both can follow vomiting. Mallory-Weiss is mucosal bleeding; Boerhaave is transmural rupture with mediastinitis risk. Severe pain, fever, tachycardia, pleural effusion, or subcutaneous emphysema should trigger emergency management.
4. **Using *H. pylori* serology as proof of active infection.** Serology can remain positive after eradication. Stool antigen and urea breath testing better identify active infection when medication washout is respected.
5. **Sending an unstable GI bleed patient to endoscopy before resuscitation.** Endoscopy treats bleeding, but airway, circulation, blood access, and variceal suspicion must be handled first.
6. **Assuming all epigastric pain is ulcer disease.** Gallbladder disease, pancreatitis, myocardial ischemia, malignancy, functional dyspepsia, and medication toxicity can mimic gastritis or PUD.

Practice questions

1. A 62-year-old man with tobacco exposure has 4 months of progressive dysphagia, first to meat and now to liquids, with 18-lb weight loss. What is the best next diagnostic step?

Upper endoscopy with biopsy. The pattern is progressive mechanical obstruction with alarm features. Barium can characterize anatomy, but tissue diagnosis is required.

2. A 38-year-old woman has intermittent dysphagia to solids and liquids plus regurgitation of undigested food at night. Barium suggests distal tapering. What test confirms the physiology?

High-resolution esophageal manometry. Achalasia requires demonstration of impaired LES relaxation and aperistalsis; endoscopy is also used to exclude pseudoachalasia.

3. A patient vomits repeatedly after heavy alcohol use and then has streaks of bright red blood in emesis but minimal pain and stable vital signs. What is the likely lesion?

Mallory-Weiss tear. It is a mucosal tear near the gastroesophageal junction after retching and often stops spontaneously.

4. A patient develops severe chest pain, tachycardia, fever, and left pleural effusion after forceful vomiting. What is the emergency diagnosis?

Boerhaave syndrome. Treat with NPO status, fluids, broad antibiotics, and urgent surgical/endoscopic source control.

5. A patient with uncomplicated dyspepsia has a positive *H. pylori* stool antigen. What is a key principle after therapy?

Confirm eradication with an active test after therapy, respecting washout: usually off antibiotics/bismuth for about 4 weeks and off PPIs for about 2 weeks when feasible.

6. A hospitalized patient with melena is hemodynamically stable and has hemoglobin 7.1 g/dL. What threshold often guides transfusion in stable nonvariceal UGIB?

A restrictive threshold around 7 g/dL is commonly used, with individualization for active ischemia, severe ongoing bleeding, or shock.

Graph-RAG extraction map

Suggested node classes: symptom, syndrome, disease, anatomic location, risk factor, diagnostic test, treatment, complication, and emergency condition. Suggested edge classes: causes, suggests, requires evaluation by, confirms, treats, complicates, mimics, and contraindicates.

Source node	Edge	Target node	Rationale phrase
Progressive solid-food dysphagia	suggests	Mechanical obstruction	Solids fail before liquids when the lumen narrows.
Dysphagia to solids and liquids from onset	suggests	Motility disorder	Transport failure affects bolus types early.
Chronic GERD	increases risk of	Barrett esophagus	Reflux injury promotes intestinal metaplasia.
Barrett esophagus	increases risk of	Esophageal adenocarcinoma	Metaplasia-dysplasia-carcinoma sequence.
Tobacco and alcohol	increase risk of	Esophageal squamous cell carcinoma	Exposure injures squamous mucosa and promotes malignancy.
Achalasia	can cause	Aspiration pneumonia	Retained food regurgitates, especially supine.
H. pylori	causes	Peptic ulcer disease	Inflammation and altered acid/mucosal defense lead to ulcers.
NSAIDs	cause	PUD and gastritis	Prostaglandin inhibition weakens mucosal protection.
Forceful vomiting	causes	Mallory-Weiss tear or Boerhaave syndrome	Pressure injury can be mucosal or transmural.
Hematemesis/melena	requires	Upper GI bleed stabilization	Airway and hemodynamics precede endoscopy.
Peptic ulcer	complicates as	Bleeding/perforation/obstruction	Ulcers can erode vessels, penetrate wall, or scar the outlet.

Selected references used for clinical cross-checking

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- Uploaded `gi1.pdf` was used only as a coverage checklist for esophageal and stomach topics, not as copied text or reproduced media.